Leonor N. L. Simões

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RESEARCH SUMMARY

Second-year PhD candidate in Astrophysics at University College London, working on the intrinsic alignments of galaxies using data from Euclid and DESI alongside cosmological simulations. My research sits at the intersection of observational cosmology, galaxy formation, and large-scale structure, with a focus on how intrinsic alignments can be used both as a cosmological probe and as a window into galaxy evolution.

EDUCATION

PhD in Astrophysics 2024 – Present
University College London • London, United Kingdom
Supervised by Prof. Benjamin Joachimi. Working on intrinsic alignments of galaxies using data from the Euclid telescope and the DESI survey.

MSci in Astrophysics 2020 – 2024
University College London • London, United Kingdom
Completed with first-class honours.

RESEARCH EXPERIENCE

Euclid SGS Support Scientist Jun 2024 – Sep 2024
UCL Astrophysics Group • London, United Kingdom
Joined the Euclid team within the UCL Astrophysics Group over the summer.

MSci Research Project Sep 2023 – Jun 2024
University College London • London, United Kingdom
Explored the properties of the cosmic web in Python, investigating how it differs between cosmological models using N-body simulations, Hessian-based cosmic web classification, and the Minimum Spanning Tree. Supervised by Dr Krishna Naidoo and Prof. Benjamin Joachimi. Awarded first-class honours for both the written report and the presentation.

Brian Duff Undergraduate Summer Research Studentship Jun 2023 – Sep 2023
University College London • London, United Kingdom
Computational project on the relationship between the Minimum Spanning Tree and the cosmic web, supervised by Dr Krishna Naidoo and Prof. Ofer Lahav. Culminated in an in-person presentation.

AstroCamp 2019 Aug 2019
Centre for Astrophysics of the University of Porto • Porto, Portugal
Selective two-week astrophysics summer programme (in English) for prospective astrophysics undergraduates. Completed a Python-based research project analysing supernova data to motivate the existence of dark energy.

PUBLICATIONS

- *"How massive neutrinos reshape the cosmic web,"* Simões L. N. L., et al., 2026, MNRAS, 549, stag1036

TALKS

- *"Implementing Three-Dimensional Intrinsic Alignment Models in Projected Simulations",* SBI for galaxy evolution, Cambridge UK, June 2026

- *"Field-level implementation of the TATT model: projection effects, smoothing choices, and the NLA limit",* Intrinsic Alignment Workshop, Institut d'Astrophysique de Paris, June 2026
- *“Probing Neutrino Mass using the Cosmic Web”*, COLOURS Summer School & Workshop, Institut Pascal (Université Paris-Saclay), Orsay, France, June 2025.

TEACHING

- **Practical Astronomy 1** (2025/26) — Third-year module on observational astrophysics at the UCL Observatory.
- **Physical Cosmology** (2025/26) — Third-year module introducing cosmological principles and selected topics in extragalactic astronomy.

AWARDS & HONOURS

- **Herschel Prize** (2024) — Best performance by a fourth-year Astrophysics student.
- **Clockmakers’ Prize** (2023) — UCL Observatory Student of the Year.
- **Brian Duff Astro Summer Research Studentship** (2023) — Competitive bursary for undergraduate summer research at UCL.
- **Huggins Prize** (2022) — Best performance by a second-year Astrophysics student.

TECHNICAL SKILLS

Programming: Python (scientific computing, data analysis). **Methods:** N-body simulations, cosmic web classification, Minimum Spanning Tree, statistical analysis of large-scale structure.